DO TEMPORAL SKILLS REALLY REPAIR AGENTS? AN INTERVENTION AUDIT OF REPAIR AND ATTRIBUTION

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Paper under double-blind review

ABSTRACT

When an agent repeatedly fails on evolving data, a reusable "skill" can appear to fix the problem even when the comparison itself is misleading. We study how to audit that claim rather than how to learn another skill. Our intervention ladder separates net repair relative to the original agent (T−N), a same-surface operation-output contrast (T−G_0), the placebo/surface perturbation itself (G_0–N), and the effect of exposing the exact same operation output through callable state rather than ordinary context (T−R_{surf}). We do not collapse T−G_0 into a pure operation effect when G_0 differs from N. This distinction changes scientific verdicts. On one DeepSeek cutoff cell, T=100% and G=72% but N=100%, so the apparent +28-point two-arm gain is no repair. Fresh DeepSeek grounding remains directionally positive under the stronger controls (+30 and +20 points for T−N and T−G_0 in two five-endpoint cells with zero observed G_0–N shift), but those cells are low-resolution. A post-hoc EIA compatibility probe illustrates why the decomposition matters: T−N is +37.5 points while G_0–N is −25 points, so we classify it as net repair with surface interaction rather than confirmation. A Kimi grounding support cell is also downgraded. Finally, R_{surf} materializes the byte-identical targeted-operation output in context. Across 18 pre-frozen DeepSeek endpoints T−R_{surf} is 0.0 points; the paired TOST passes at the prospectively frozen ±10-point portfolio margin, with paired-t 95% CI $[-8.5, 8.5]$ and bootstrap 95% sensitivity CI $[-8.3, 8.3]$. Because an observed-variance sensitivity calculation is only about 53%, the portfolio is ceiling-heavy, and the strictly non-ceiling subset is unresolved, we treat this only as a low-sensitivity frozen-margin surface check. Thus stronger controls can preserve bounded net repair, expose surface interaction, reverse support cells, and remove skill-container credit within the tested harness without implying broad equivalence to temporal-retrieval algorithms. The contribution is the audit and the verdict changes it forces.

1 INTRODUCTION

Suppose an agent is asked: *What was the latest value known by the end of April?* A May release is topically relevant, easy to retrieve, and wrong for the question. On the next period the numbers change, but the required operation does not: filter evidence by the information cutoff. Similar recurring operations arise when an agent must distinguish publication time from reporting time or ground a numerical change in documentary context available at the time. Reusable skills are an appealing way to persist such procedures, but a successful skill run does not by itself identify what was repaired or what part of the implementation deserves credit.

The first problem is the comparator. A targeted helper can beat a generic helper simply because the generic helper degrades the original agent. We observe exactly this pattern on a frozen DeepSeek cutoff cell: targeted T succeeds on 100%, target-blind G on 72%, and the original agent N on 100% of endpoint-repeat observations. A conventional T-versus-G evaluation reports +28 points; adding N changes the verdict to *no repair*. This is not a variance correction. It changes which scientific claim is admissible.

The second problem is attribution. We therefore keep four observable contrasts separate:

$$\underbrace{\Delta_{T-N}}_{\text{net repair}}, \quad \underbrace{\Delta_{T-G_0}}_{\text{same-surface output contrast}}, \quad \underbrace{\Delta_{G_0-N}}_{\text{surface/placebo perturbation}}, \quad \underbrace{\Delta_{T-R_{\text{surf}}}}_{\text{integration-surface effect}}. \tag{1}$$

G_0 is an information-empty no-operation placebo exposed through the same helper surface as T. T–G_0 therefore asks whether the targeted operation output changes behavior *conditional on that surface*; G_0–N separately reveals whether the surface/placebo itself perturbs the original agent. We do not call T–G_0 a pure operation effect when G_0–N is nonzero. Algebraically, T–N=(T–G_0)+(G_0–N), so the three contrasts expose rather than hide a surface interaction. R_{surf} asks a different question: it runs the identical frozen operation on identical candidate evidence but materializes the exact output in ordinary context instead of callable state. It is deliberately a controlled surface intervention, not an independently learned temporal retriever.

This audit changes several interpretations. Fresh DeepSeek grounding runs have +30 and +20 point T–N and T–G_0 directions in C3/C4 with zero observed G_0–N shift on those runs, although each cell has only five endpoints. In a post-hoc EIA compatibility probe, T–N is +37.5 points but G_0–N is -25 points; we use this cell primarily to illustrate net repair with surface interaction, not as confirmation. A Kimi same-domain grounding support cell is downgraded under the fresh controls. The surface audit narrows credit again: across 18 pre-frozen DeepSeek load-bearing endpoints T–R_{surf} averages 0.0 points. A formal two-one-sided equivalence test at the prospectively frozen ±10-point portfolio margin passes; paired-t and bootstrap 95% intervals are $[-8.5, 8.5]$ and $[-8.3, 8.3]$ points, respectively. Because a post-hoc observed-variance sensitivity calculation is only about 53%, the portfolio is ceiling-heavy, and the strictly non-ceiling subset is too small, we use this only as a low-sensitivity frozen-margin surface check rather than a general equivalence result.

Our setting is a source-native replay built from first-party U.S. Bureau of Economic Analysis (BEA), NOAA Climate Prediction Center, and Energy Information Administration (EIA) release histories (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026; U.S. Energy Information Administration, 2026). The ledger contains 35 frozen endpoints and 2,056 retained runtime-valid rows spanning cutoff validity, release alignment, and exogenous grounding. The initial BEA/NOAA object is motivated by TimeSage-EV but is not an exact rerun because the evaluated TimeSage-EV snapshot remains unavailable (Yao et al., 2026). EIA endpoint construction, the 4/8 temporal split, scorer, procedure hashes, and condition order were frozen before EIA outcomes, but EIA itself was selected post-review because its recurring dated releases are structurally compatible with the cutoff mechanism. We therefore treat it as a post-hoc compatibility probe, not a prospective cross-domain confirmation. Endpoints, not repeated executions, are the independent scientific units.

The paper is an evaluation and identification study, not a skill-learning algorithm. Prior work already studies causal skill attribution, reusable-skill evolution, and temporal retrieval (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026; Asai et al., 2024; Yan et al., 2024; Han et al., 2025; Wang et al., 2026a). We do not claim novelty in procedure reuse, time-aware retrieval, or paired skill causality. The residual contribution is an audit contract for assigning scientific credit and an empirical demonstration that stronger controls can reverse a repair verdict, expose a surface interaction, downgrade a support cell, and erase container-level credit while leaving some net repairs intact.

Contributions. We make three contributions. First, we separate net repair, same-surface operation-output contrast, placebo/surface perturbation, and integration-surface effect rather than collapsing them into one "skill effect." Second, we show that this separation changes conclusions: T>G but T=N is an observed false positive; fresh controls preserve bounded DeepSeek repairs while reclassifying EIA as surface-interacting and downgrading Kimi grounding. Third, an exact-output context-materialization control yields a passing TOST at the prospectively frozen ±10-point *portfolio-average* margin; because the design has low sensitivity and a ceiling-heavy portfolio, we treat this as a frozen-margin surface check while leaving the low-count non-ceiling subset and broader temporal-retrieval questions unresolved. The contribution is therefore methodological understanding: within the registered endpoint portfolio and tested resolution, stronger controls can preserve, narrow, or reverse an apparent reusable-skill effect.

2 An Intervention Audit of Reusable Procedures

The audit treats a reusable procedure as an intervention rather than a label attached to a successful run. Let endpoint i specify a question x_i, evidence package E_i, model configuration M_i, and familyspecific binary outcome $Y_i \in \{0,1\}$. For repeated executions under arm a, let $\bar{Y}_i(a)$ be the repeat mean. Repeats measure execution robustness; endpoints remain the independent scientific units.

Base stress test: N, G, and T. The historical design contains the original agent N, a complexity-matched target-blind *stress helper* G, and the targeted procedure T. G is intentionally mechanism-misaligned rather than a benign or optimal generic assistant; its role is to test whether T-versus-structured-assistance can produce a misleading verdict. We report

$$\widehat{\Delta}_{T-N} = \frac{1}{|\mathcal{I}|}\sum_i \big(\bar{Y}_i(T) - \bar{Y}_i(N)\big), \qquad \widehat{\Delta}_{T-G} = \frac{1}{|\mathcal{I}|}\sum_i \big(\bar{Y}_i(T) - \bar{Y}_i(G)\big). \tag{2}$$

T–N is the net effect of the targeted intervention relative to the original agent. T–G asks whether it beats the executed target-blind helper. G is not assumed benign, so T>G with T≤N is no repair.

Same-surface placebo: $\mathbf{G_0}$. G_0 executes the same helper-exposure path as T but is information-empty: it reads neither package nor context and returns an empty object. We report

$$\widehat{\Delta}_{T-G_0} = \frac{1}{|\mathcal{I}|}\sum_i \big(\bar{Y}_i(T) - \bar{Y}_i(G_0)\big), \qquad \widehat{\Delta}_{G_0-N} = \frac{1}{|\mathcal{I}|}\sum_i \big(\bar{Y}_i(G_0) - \bar{Y}_i(N)\big). \tag{3}$$

T–G_0 is a *same-surface operation-output contrast*: it asks whether supplying the targeted output changes behavior relative to no operation output under the same exposure surface. G_0–N is the corresponding surface/placebo perturbation diagnostic. These quantities obey the observed-mean identity

$$\widehat{\Delta}_{T-N} = \widehat{\Delta}_{T-G_0} + \widehat{\Delta}_{G_0-N}. \tag{4}$$

Equation 4 is a decomposition of contrasts, not a mediation theorem. When G_0–N is nonzero, T–G_0 is conditional on a perturbed surface and must not be called a pure operation effect. Accordingly, we classify T–N>0 as *net repair*; T–G_0 is reported separately as evidence about the output under that surface. Cells with nonzero placebo shift are explicitly labeled surface-interacting rather than operation-attributed.

Integration-surface control: $\mathbf{R_{surf}}$. R_{surf} runs the exact same frozen operation source on the exact same candidate evidence as T, then materializes the identical operation output inside ordinary context while exposing no callable helper. Thus

$$\widehat{\Delta}_{T-R_{\text{surf}}} = \frac{1}{|\mathcal{I}|}\sum_i \big(\bar{Y}_i(T) - \bar{Y}_i(R_{\text{surf}})\big) \tag{5}$$

is an integration-surface estimand under the frozen forced one-answer harness. R_{surf} is deliberately not an independently learned or heuristic temporal retriever. A zero-call parity audit verifies that candidate evidence is preserved, operation-output content is identical, and only where the precomputed output enters the harness changes. The prospectively frozen material threshold is 10 percentage points on the pre-frozen DeepSeek endpoint portfolio. Multi-turn deployments in which a callable helper can be invoked adaptively require a different control.

The audit targets three recurring failure families. **Cutoff validity** filters evidence by the target information cutoff. **Release alignment** maps a release to the reporting period represented by its content. **Exogenous grounding** links a numerical change to first-party documentary evidence available before the cutoff. No targeted procedure contains endpoint-specific answers, gold labels, or fitted values; procedure hashes are frozen before execution and reused on held-out releases.

Three falsifiers follow. First, T>G with T≤N is no repair. Second, a positive T–G_0 does not by itself establish a pure operation effect when G_0–N is nonzero; all three contrasts in Eq. 4 must remain visible. Third, if the integration surface matters under the forced one-answer harness, T should leave a material residual over R_{surf} when operation output and evidence are held exactly fixed.

3 Related Work

Causal skill attribution and agent evaluation. Paired trace auditing, no-skill comparisons, randomized skill masking, and continual skill evaluation already show that exposing an agent to a skill

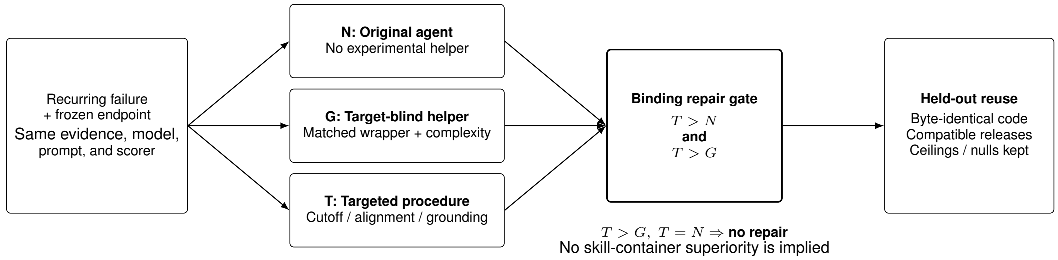

Figure 1: Base three-arm stress test. N anchors the original agent, G is a target-blind mechanism-misaligned stress helper, and T contains the targeted temporal operation. G_0 exposes placebo/surface perturbation; R_{surf} isolates where the same precomputed operation output enters the forced one-answer harness.

can change behavior and that effects are heterogeneous (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026). We therefore do not claim paired skill causality or heterogeneous effects as new. Our question is narrower and evaluative: does the targeted intervention improve the original agent, does the target operation output matter under the same exposure surface, and does moving the same precomputed output between two integration surfaces change the final answer under a forced one-answer harness? The observed T>G but T=N cutoff cell demonstrates why these are different estimands rather than interchangeable baselines.

Skill evolution and reusable procedures. Evo-Harness, SkillProx, HyperSkill, Anything2Skill, and related systems study continual maintenance, compilation, optimization, structured skill memory, and conversion of external knowledge into reusable procedures (Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026; Pan et al., 2026). These works make procedure reuse and skill accumulation established objects. We treat acquisition as upstream: after a reusable procedure exists, our contribution is an audit of the scientific credit assigned to it. A stronger control may preserve a net repair while exposing a surface interaction, eliminating a skill-container interpretation, or downgrading a previously positive support cell.

Temporal retrieval and context construction. Selective/corrective RAG and temporal retrieval already address irrelevant evidence, time-sensitive indexing, and explicit temporal constraints upstream of generation (Asai et al., 2024; Yan et al., 2024; Zhang et al., 2025; Abdallah et al., 2026; Han et al., 2025; Wang et al., 2026a; Liu et al., 2026). Consequently, a useful cutoff or alignment operation need not be packaged as callable state. Our R_{surf} experiment does *not* compare T with an independently learned temporal retriever. It intentionally runs the same frozen operation and materializes the identical output in context, creating a same-information surface-isolation test. The result therefore speaks only to integration-surface placement under the frozen one-answer harness, not to algorithmic equivalence between skills and temporal RAG or to adaptive callable use in deployment. A broader retrieval claim would require a separately implemented retriever or reranker that must discover the relevant transformation from the same raw candidates; we do not make that claim here.

The residual novelty is thus methodological: a reusable-skill evaluation should distinguish net repair, same-surface output contrast, placebo/surface perturbation, and integration-surface value before assigning credit to “the skill.” In our data, making those distinctions changes multiple verdicts rather than merely adding confidence intervals to an unchanged story.

4 Interventions and Audit Contracts

All arms replay the same frozen endpoint with the same evidence package, model configuration, output schema, and family scorer. We intervene only on the procedure exposure or on where an already computed procedure output enters the harness. The scientific unit is the endpoint; repeated executions are robustness observations rather than additional tasks.

4.1 Base N/G/T stress test

N adds no experimental helper. G exposes a frozen target-blind *stress helper* using the same callable surface and matched implementation complexity as T, but it is forbidden from performing the registered temporal operation. G is not presented as a benign, optimal, or representative generic assistant: its purpose is to stress the common T-versus-generic evaluation pattern and reveal when control-arm degradation changes the verdict. For cutoff validity, G can validate package structure but cannot compare release dates with the cutoff; for release alignment, it cannot map publication dates to reporting periods; for grounding, it cannot perform the mechanism-specific contextual selector. Targeted and G source implementations are matched in lexical-token and AST-node count for each family.

This base test is useful precisely because G need not be benign. T–G is a stress contrast against structured but mechanism-misaligned assistance, while T–N determines whether anything was actually repaired. We do not use T–G alone to estimate a generic-helper treatment effect. The DeepSeek cutoff false positive in Section 6 shows why the N anchor is mandatory whenever such a stress comparator is used.

4.2 Same-surface no-operation placebo G_0

G_0 executes the same assigned-helper path as T but its frozen function is simply `def skill(package, context): return {}`. It reads neither evidence nor task context. Consequently T and G_0 differ in whether the target operation output is present while sharing the same helper-exposure surface. We do not label G_0 behaviorally neutral at the stratum level. Its empirical difference from N is reported directly.

The final audit therefore does not combine T–N and T–G_0 into a single pure-operation estimand. T–N carries the *net-repair* claim. T–G_0 is a *same-surface operation-output contrast*; G_0–N reports how much the placebo/surface itself perturbs N. Their observed means satisfy Eq. 4. When G_0 harms or helps N, the T–G_0 contrast is conditional on that perturbed surface and is reported as such. The separately frozen A1/A2 G_0 studies remain diagnostics of wrapper perturbation; their full-track equivalence result is not transported into every small load-bearing stratum.

4.3 Context-materialization surface control R_{surf}

R_{surf} isolates a narrower implementation question. It runs the exact targeted-operation source on the same candidates and computes the exact same operation output as T. Instead of exposing that output as `reusable_helper_output`, it adds the object as `retrieval_operation_output` inside the ordinary evidence context and exposes no callable helper. A zero-call audit verifies for every tested endpoint that removing this one added field reconstructs the original evidence package, operation-output hashes match T exactly, and candidate evidence is neither removed nor reordered.

Because R_{surf} deliberately reuses T's operation output, it is a *surface-isolation control*, not an independent temporal-retrieval algorithm. It asks whether placing the same precomputed output on the T surface rather than in ordinary context changes the final answer under this forced one-answer harness. It does not test adaptive callable use during generation, whether a learned temporal retriever can discover or approximate the operation, or whether persistent callable state has an advantage in multi-turn deployment.

4.4 Frozen outcomes, thresholds, and evidence layers

Before each source-native execution tranche, endpoint identity, family outcome, repeat count, condition order, model configuration, and procedure source hashes are frozen. A cutoff success requires a nonempty cited set and latest selected reference that are all available by the target cutoff. A release-alignment success requires the correct reporting-period identity and aligned value. A grounding success requires the requested evidence slots to be filled by frozen first-party sentence IDs in the correct order. Auxiliary stricter scorers are reported only as sensitivity analyses.

The original R_{surf} contract freezes a 10-point material effect threshold on the 18-endpoint DeepSeek portfolio, before R_{surf} outcomes. The margin is conservative relative to the already observed 20–60 point scale of the load-bearing targeted effects available when the surface experiment was designed. We report the preregistered 90% endpoint-bootstrap interval, a post-hoc 95% bootstrap sensitivity interval, and a paired-endpoint TOST at the unchanged ± 10-point margin. The TOST is a statistical sensitivity analysis of the same frozen decision boundary rather than a replacement gate. For G_0, A1 and A2 remain unpooled; exact-stratum G_0–N diagnostics show where full-track equivalence does and does not transport.

The initial BEA/NOAA replay freezes pre-April and April–May periods before outcomes. A separate post-hoc EIA compatibility probe uses 16 fixed weekly releases to construct 12 deterministic cutoff endpoints: source indices 3–14, with four releases before April 1 assigned to C3-R4 and eight later releases assigned to C4-R4. Endpoint construction, scorer, T1/G1 hashes, and randomized condition order were frozen before EIA model outcomes. However, EIA was selected post-review because its dated weekly releases are structurally compatible with the cutoff mechanism, so C4-R4 is an illustrative mechanism-compatibility probe rather than prospective cross-domain confirmation.

5 Experimental Setup: Source-Native Replay

TimeSage-EV motivates the three recurring mechanism families—cutoff validity, release alignment, and exogenous grounding—and exposes persistent skills to later periods (Yao et al., 2026). Our experiment is not an exact rerun of its unavailable evaluated snapshot. Instead, we reconstruct content-addressed endpoints directly from first-party institutional release histories and intervene on the same reusable operations.

5.1 Frozen endpoints and scorers

The initial replay contains 23 endpoints from BEA GDP releases and NOAA-CPC ENSO Diagnostic Discussions (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026): twelve pre-April endpoints form C3-R and eleven April–May endpoints form held-out C4-R. Raw pages are SHA-256 bound, evidence identifiers are opaque, and grounding exposes the full content-region sentence pool rather than a gold-only candidate set. Family scorers are deterministic and frozen before model outcomes; a pre-execution self-test gives 23/23 success on gold outputs and 0/23 on constructed negatives.

A post-review compatibility probe adds 12 EIA Weekly Petroleum Status Report cutoff endpoints from a third institutional system (U.S. Energy Information Administration, 2026). EIA is chosen because its recurring first-party releases and explicit publication dates fit the already frozen cutoff interface. Four endpoints form an earlier diagnostic split and eight April–May endpoints form the later probe split. T1 and G1 are reused byte-for-byte; source manifest, endpoints, scorer, harness, and condition order are frozen before execution. Because the institution itself was selected post-review for mechanism compatibility, EIA is an illustrative compatibility probe rather than independently preregistered cross-domain confirmation.

5.2 Questions and inference contract

RQ1 asks whether adding the original-agent anchor changes a verdict that would look positive under targeted-versus-generic comparison alone. RQ2 asks whether the targeted operation output improves on a same-surface no-operation placebo G_0 while the complete targeted intervention also improves on N. G_0–N is reported separately as a surface-perturbation diagnostic rather than assumed neutral within every stratum. RQ3 asks whether byte-identical procedures retain direction on compatible held-out releases while preserving ceiling and incompatible-domain nulls. RQ4 asks whether moving the exact same frozen temporal-operation output between the T surface and ordinary context leaves a material residual under the forced one-answer harness.

The endpoint is the independent scientific unit. Repeats characterize execution robustness and are averaged within endpoint; they are never counted as additional tasks. For non-ceiling cells we report endpoint win/tie/loss patterns, percentile bootstrap intervals over endpoints, and exact two-sided sign tests over non-ties when resolution permits. Two- and three-endpoint cells are explicitly descriptive. We do not pool heterogeneous procedures, models, or domains into an omnibus procedure-effect statistic.

5.3 Evidence layers and parity

DeepSeek-V4-Pro is the registered primary track with five-repeat historical replay plus fresh identification runs. A pre-outcome support rule selects Kimi-K3 as a separately labeled replication; a later hash audit removes one overwritten historical repeat block without replacement. Doubao-Seed-2.0-Mini is an exploratory capacity stress. The EIA layer evaluates DeepSeek and the mini model. After adding the prospectively frozen G_0 and R controls, the retained source-native evidence ledger contains 2,056 runtime-valid rows over the same 35 distinct endpoints from BEA, NOAA, and EIA. The endpoint count, not the enlarged repeat/control-row count, determines scientific resolution.

Every targeted procedure implementation has a family-matched generic helper with the same callable signature, identical Python lexical-token count and AST-node count, and equal external tool-call budget. G_0 uses the same helper-execution surface but is information-empty: it returns $\{\}$ and reads neither package nor context. For R_{surf}, zero-call parity verifies on every tested endpoint that removing the added context field recovers the original evidence package exactly and that the materialized operation output is content-identical to T's helper output. This makes R_{surf} a controlled integration-surface intervention, not an independently designed retriever. Full execution chronology, content hashes, per-call CSV checkpoints, and provenance bindings are reported in Appendix A and the supplement.

6 Results

We report endpoint-level contrasts and keep repeated executions inside each endpoint mean. The results are organized by what each added control changes about the scientific verdict rather than by a universal "skills help" narrative.

6.1 The original-agent anchor overturns a two-arm gain

On DeepSeek C3-R cutoff, T succeeds on 100% of endpoint-repeat observations and the complexity-matched target-blind stress helper G on 72%. A conventional T-versus-G comparison therefore reports +28 points. N is also 100%, however, so T–N=0 and the cell is *no repair*. At endpoint means, T–G has 4 wins/1 tie/0 losses, whereas T–N has 0 wins/5 ties/0 losses, so the verdict reversal does not depend on treating repeats as independent tasks. A zero-call audit further confirms equal helper invocation and no detectable condition-position association; T prompts are 18.85 input tokens shorter on average. The point is not that G is a universally fair generic assistant, but that T-versus-stress-helper evidence can look convincingly positive while the original agent was never repaired.

6.2 G_0 is a placebo diagnostic, not a universal neutrality assumption

The independently frozen A2 study shows that G_0 is close to N when averaged over the full DeepSeek endpoint portfolio (G_0–N=-2.1 points, 90% CI $[-9.3, 5.0]$); Kimi is similarly close on its full endpoint set (+2.2 points, $[-4.3, 8.7]$). These global summaries do *not* transport into every small load-bearing stratum. On the exact A2 C3 and C4 grounding strata, G_0–N is +5 points and the 90% intervals extend to +15 points; on the eight EIA held-out endpoints it is -18.8 points with a wide interval. We therefore remove stratum-level "behaviorally neutral" language and use G_0 as a same-surface no-operation placebo.

The three contrasts are more informative when kept separate. In a fresh two-repeat tranche, DeepSeek C3 grounding has T–N=+30, T–G_0=+30, and G_0–N=0 points (2 endpoint wins, 3 ties, 0 losses for the first two contrasts); C4 grounding gives +20, +20, and 0 (1 win, 4 ties, 0 losses). The endpoint-bootstrap intervals include zero because each cell has five endpoints. These are directional net-repair and same-surface output results with zero *observed* placebo shift, not formal stratum-level equivalence or pure-operation identification.

The EIA cell illustrates the surface interaction directly. On eight post-hoc compatibility-probe endpoints, fresh T–N is +37.5 points with 90% bootstrap interval $[18.8, 56.3]$, five wins, three ties, no losses, and one-sided exact sign test $p = 0.03125$. T–G_0 is +62.5 points while G_0–N is -25 points, giving the exact contrast decomposition $+37.5 = +62.5 - 25$. We classify this as *net repair with surface interaction*, not confirmatory evidence: T improves on N, but T–G_0 is conditional on a

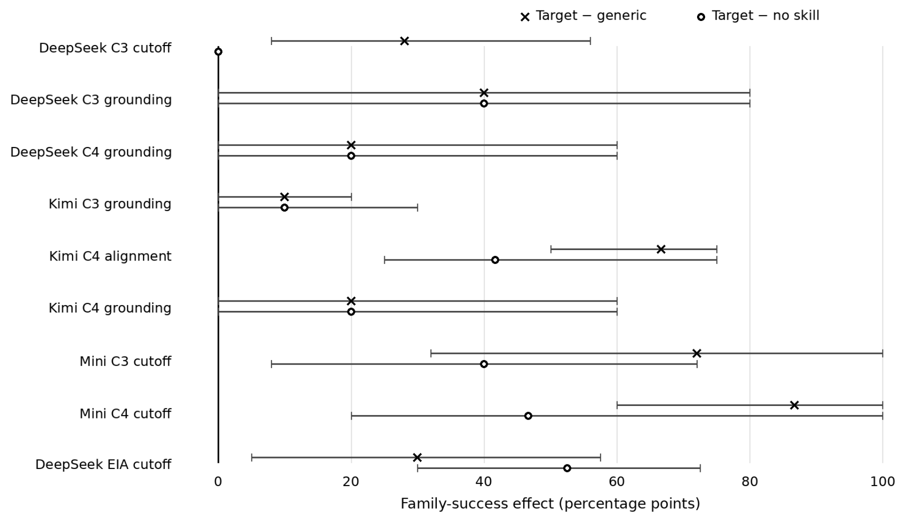

Figure 2: Temporal-operation effects are regime-dependent. Circles compare T with N; crosses compare T with historical target-blind G. DeepSeek C3 cutoff illustrates the identification point: T>G but T=N, so the apparent two-arm gain is not a repair. EIA is a post-hoc compatibility probe on a separate institutional endpoint set.

harmful placebo surface. The earlier five-repeat EIA execution is directionally consistent, but both executions reuse the same endpoint set and are not independent endpoint samples.

6.3 Grounding survives only as a bounded DeepSeek result

The historical DeepSeek grounding cells are 20/20/60 for N/G/T in C3 and 40/40/60 in C4. Fresh N/G_0/T execution preserves +30 and +20 point net-repair and same-surface output directions with zero observed G_0–N shift in those tranches. Kimi does not supply cross-model grounding confirmation: its fresh same-domain grounding check has T–$G_0 = -25$ points and T–N=0, and cross-domain grounding is also null. We therefore drop the earlier cross-model grounding-transfer interpretation and retain only a bounded DeepSeek grounding result.

6.4 EIA is a post-hoc compatibility probe

The EIA expansion is constructed from 16 fixed weekly releases. Source indices 3–14 deterministically define 12 endpoints; releases before April 1 form four earlier diagnostics and the remaining eight form C4-R4. The endpoint set, time split, scorer, T1/G1 hashes, and randomized condition order are frozen before EIA outcomes. But EIA itself was chosen post-review because its dated releases fit the already frozen cutoff interface. We therefore use it only as a mechanism-compatibility illustration, not as confirmatory or generic cross-domain evidence. The mini model is already at N ceiling on the same eight endpoints, further emphasizing regime dependence.

6.5 Exact-output control passes a low-sensitivity frozen-margin check

R_{surf} deliberately computes the exact T operation output and changes only how that output enters the harness. Across 18 pre-frozen DeepSeek endpoints, T–R_{surf} is 0.0 points (1 win/16 ties/1 loss). The preregistered 90% bootstrap interval is $[-5.6, 5.6]$; a paired TOST at the unchanged $\pm$10-point margin passes (both one-sided $p = 0.0121$), with paired-t 95% CI $[-8.5, 8.5]$ and bootstrap 95% sensitivity CI $[-8.3, 8.3]$. We treat this as a *low-sensitivity frozen-margin check*, not positive evidence that the surfaces are generally interchangeable: an observed-variance sensitivity calculation is only about 53% and the portfolio is ceiling-heavy.

The zero mean is not produced only by ceiling cells: C3 grounding is 60%/60% under T/R_{surf}, and C4 is 70%/70%. However, the four strictly non-ceiling endpoints have a wide 90% interval $[-25, 25]$, so non-ceiling equivalence remains unresolved; EIA is ceiling at 100%/100%. A three-endpoint Kimi alignment tie is descriptive only. Because R_{surf} reuses T's exact output under a forced one-answer harness, no multi-turn callable-state or independent temporal-retrieval equivalence is implied.

7 Limitations

The frozen TimeSage-EV evaluated snapshot remains unavailable, so the 35-endpoint BEA/NOAA/EIA study is independent source-native evidence rather than an exact rerun. Resolution is uneven: the main grounding cells have five endpoints, Kimi alignment three, Kimi same-domain grounding two, and cross-domain grounding is null. G_0 is an information-empty same-surface placebo, not a universally neutral control: full-track equivalence does not transport to every stratum, and EIA G_0 underperforms N. Accordingly, T–N carries net repair while T–G_0 and G_0–N remain separate output/surface contrasts.

R_{surf} tests only placement of the exact same precomputed output under a forced one-answer harness. The portfolio TOST passes at the frozen ± 10-point margin, but sensitivity is about 53%, the portfolio is ceiling-heavy, and four strictly non-ceiling endpoints remain unresolved; we therefore treat it as a low-sensitivity margin check, not positive evidence of general surface equivalence. Adaptive callable use and independent temporal retrievers are untested. EIA is a post-hoc compatibility probe: its endpoint construction and temporal split were outcome-independent, but the institution was not independently sampled, so it does not support prospective cross-domain confirmation or model-invariant transfer.

8 Conclusion

A reusable "skill effect" can contain several different causal stories. In our audit, the original-agent anchor turns one apparent +28-point targeted-versus-stress-helper benefit into no repair; a same-surface no-operation placebo exposes output and surface effects without requiring neutrality; and an exact-output context-materialization control makes the observed portfolio consistent with no integration-surface difference larger than the frozen ± 10-point margin under the forced one-answer harness. Stronger controls also force a Kimi grounding downgrade while leaving only bounded, low-resolution positive repair evidence.

The resulting contribution is therefore not that temporal skills are generally superior. It is an intervention audit for assigning credit: use T–N for net repair; keep T–G_0 and G_0–N separate to expose same-surface output and placebo/surface effects; and use T–R_{surf} only as an integration-surface test when the same precomputed output is placed on two surfaces under the same one-answer harness. In this study, stronger controls preserve some bounded net repairs, expose a surface interaction, reverse support cells, and produce a passing but low-sensitivity frozen-margin surface check. Broader claims about non-ceiling equivalence, adaptive callable interfaces, or temporal retrieval remain outside the evidence.

Reproducibility Statement

The source-native study is content-addressed end to end. The supplementary replication bundle contains the frozen endpoint definitions, prompt harnesses, family scorers, all six executed targeted/generic helper implementations, condition schedules, first-party archived source pages, retained raw result JSON, adjudication code, and a SHA-256 manifest. Appendix A records the executed helper code, the endpoint-level inferential protocol, and the zero-call invocation/order/date-serialization audits. The source-native bundle is independent of the unavailable TimeSage-EV evaluated snapshot; the exact TimeSage replication remains a separately identified external replication target.

AI Use Statement

Generative AI tools were used throughout the research workflow to assist literature triage, research-idea and hypothesis refinement, experimental-design critique, implementation and debugging, data reformatting, result interpretation, internal mock review, and manuscript drafting and editing. AI-assisted experimental artifacts were accepted only after executable checks against frozen task contracts, immutable raw outputs, first-party source material, or deterministic recomputation. AI-generated review feedback had advisory status and did not itself create scientific evidence. The authors reviewed the AI-assisted work and take responsibility for the final paper, claims, code, and supplementary artifacts.

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A Reproducibility and Sensitivity Audits

T1/G1 are byte-identical across the initial replay and EIA validation. The provenance-clean r3 reaudit is `6221016166b71ebe`; r4 adjudication is `316ee89a8eb1d53f`. The r5 replication bundle contains executable r3/r4 endpoints, harnesses, scorers, runner scripts, frozen helper source, first-party archived pages, retained result JSON, and a SHA-256 manifest. This bundle reproduces the source-native study and remains separate from the unavailable TimeSage-EV evaluated snapshot.

A.1 Exact frozen reusable helpers

The six helper implementations below are the executed source, not pseudocode. Each targeted/generic pair has equal lexical-token count and equal AST-node count under the pre-execution audit.

```
# T1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("cutoff_date", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("release_date", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# G1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("schema_ceiling", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("schema_version", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# T2_release_alignment.py
import re
def skill(package, context):
    text = " ".join(package.get("document_text", "").split())
    match = re.search(r"([1-4])(?:st|nd|rd|th) Quarter(?: and Year)? (20\d{2})",
                      text, re.I)
    value = None
    if match:
        value = f"{match.group(2)}-Q{match.group(1)}"
    return {"value": value}

# G2_release_alignment.py
import re
def skill(package, context):
    text = " ".join(package.get("document_text", "").split())
    match = re.search(r"(News) (Release|Archive|Statement)(?: and Data)? "
                      r"([A-Za-z]{2,20})", text, re.I)
    value = None
    if match:
        value = f"{match.group(1)}-{match.group(3)}"
    return {"value": value}

# T3_exogenous_grounding.py
def skill(package, context):
    markers = ("contributors to", "reflected", "because", "collectively",
```

```
                "favored", "favors ", "associated with", "due to")
    chosen = []
    for span in package.get("candidate_spans", []):
        text = span.get("text", "").lower()
        sid = span.get("span_id", "")
        if any(marker in text for marker in markers):
            chosen.append(sid)
    return {"span_ids": chosen}

# G3_exogenous_grounding.py
def skill(package, context):
    markers = ("table", "figure", "release", "appendix", "contact",
               "notes ", "technical notes", "page")
    chosen = []
    for span in package.get("candidate_spans", []):
        text = span.get("text", "").lower()
        sid = span.get("span_id", "")
        if any(marker in text for marker in markers):
            chosen.append(sid)
    return {"span_ids": chosen}
```

A.2 Exploratory capacity-stress matrix

Split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
C3-R	Cutoff	32	0	72	**+72**	**+40**
C3-R	Align	100	100	100	0	0
C3-R	Ground	40	48	64	+16	+24
C4-R	Cutoff	53	13	100	**+87**	**+47**
C4-R	Align	80	93	100	+7	+20
C4-R	Ground	60	52	64	+12	+4

Table 1: Exploratory mini-model capacity stress over five repeats (percent), reported separately from the registered primary and replication tracks.

A.3 Endpoint-level uncertainty and test resolution

Table 2 averages repeats within endpoint-condition and bootstraps endpoints; W/T/L is targeted versus generic, and sign tests use non-ties. Repeats never increase the inferential sample size. For n non-tied endpoints, the smallest attainable two-sided sign-test value is 2^{1-n} when every endpoint moves in the same direction. Thus two non-tied endpoints cannot yield a value below .5; three cannot go below .25; five cannot go below .0625; eight can reach .0078. Ties reduce resolution further. This arithmetic is why the two-endpoint grounding-transfer result is explicitly descriptive, why the three-endpoint Kimi alignment result remains underpowered despite a large effect, and why the eight-endpoint EIA validation supports a stronger endpoint-level claim.

Layer	Split	Fam.	Δ_{T-G} [95%]	W/T/L	p_G	Δ_{T-N} [95%]	p_N
D	C3	Cut	28 [8,56]	4/1/0	.125	0 [0,0]	1
D	C3	Grd	40 [0,80]	2/3/0	.50	40 [0,80]	.50
D	C4	Cut	13 [0,20]	2/1/0	.50	0 [0,0]	1
D	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
K	C3	Grd	10 [0,20]	2/3/0	.50	10 [0,30]	1
K	C4	Align	66.7 [50,75]	3/0/0	.25	41.7 [25,75]	.25
K	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
M	C3	Cut	72 [32,100]	4/1/0	.125	40 [8,72]	.25
M	C3	Grd	16 [-12,60]	1/3/1	1	24 [0,64]	.50
M	C4	Cut	86.7 [60,100]	3/0/0	.25	46.7 [20,100]	.25
M	C4	Align	6.7 [0,20]	1/2/0	1	20 [0,60]	1
M	C4	Grd	12 [0,28]	2/3/0	.50	4 [-32,36]	1

Table 2: Endpoint-level uncertainty for non-ceiling initial-replay cells. D=DeepSeek primary, K=Kimi replication, M=exploratory mini.

A.4 Why the family estimand is primary

The family contract and the full-task conjunctive score answer different questions. The family score asks whether the targeted temporal operation was corrected. The conjunctive score additionally requires orthogonal output fields and exact surface forms. A zero conjunctive score can therefore coexist with a successful cutoff intervention when temporal admissibility is correct but an unrelated field differs, or with successful release alignment when a semantically identical period is formatted differently. The family definitions were frozen before execution, so selecting them is determined by the experimental question and its time order. The conjunctive score is preserved as a sensitivity view and is never used to erase an unfavorable family result.

Family	r3 full-task conjunctive score	Effect on claim
Grounding	Identical to family score	Unchanged
Cutoff	All cells collapse to 0	Family estimand remains primary
Alignment	All cells collapse to 0	Family estimand remains primary

Table 3: Sensitivity to the auxiliary full-task exact score. The two columns evaluate different estimands.

A.5 Call-rate, prompt-length, and condition-position audit

The retained source-native rows contain 442 no-skill, 442 generic, and 442 targeted observations. Helper execution is deterministic in the frozen harness: generic and targeted execute exactly once before the model answer in 442/442 rows each, whereas no-skill executes no helper in 0/442 rows. Thus targeted-minus-generic cannot be explained by a higher helper invocation rate; this experiment does not measure endogenous tool-use propensity. Across exact targeted/generic pairs, targeted prompts are shorter by 18.85 input tokens on average (302 shorter, 140 longer). Condition-position counts are 157/135/150 for no-skill, 145/155/142 for generic, and 140/152/150 for targeted across positions 0/1/2. Pooled 3×2 success-by-position tests give $p = .993$, .657, and .914 for no-skill, generic, and targeted respectively; targeted-minus-no-skill remains positive at all three positions. These are post-hoc zero-call diagnostics and do not substitute for a prospectively counterbalanced replication. The full audit is included in the supplement.

A.6 ISO date and period-representation audit

A zero-call static audit covers the 23 initial BEA/NOAA endpoints and the 12 post-review EIA endpoints. Across the serialized endpoint objects, all 55 occurrences of `cutoff_date` and all 176 occurrences of `release_date` are calendar-valid `YYYY-MM-DD` strings. Whenever both top-level and `skill_context` cutoff fields are present they agree exactly. T1 makes 88 observed release-versus-cutoff comparisons over the frozen endpoint packages; lexical string order and parsed-calendar order agree on all 88. The executed T1 source is byte-identical between the initial and EIA phases (SHA-256 `df1950ef3930...d56d522`). Reporting-period identities are a separate representation layer: the initial replay uses canonical month/quarter identities for relevant families, while EIA uses day-level data-ending dates. These reporting-period strings never enter T1's release-date comparator and are adjudicated by the frozen family scorer.

A.7 Completed identification ladder

Table 4 separates four distinct questions that are all executed on the source-native replay. The controls were added prospectively in sequence: outcome thresholds, endpoint units, integration surfaces, model identities, and checkpoint policies were frozen before the corresponding fresh outcomes.

G_0: same-surface no-operation placebo. G_0 executes exactly one assigned helper call through the same harness surface as T but is information-empty: the frozen function is `def skill(package, context): return {}`. It reads neither package nor context. The first DeepSeek 35-endpoint two-repeat tranche gives G_0–N 90% CI $[-14.3, 2.9]$ points; an independently frozen four-repeat A2 study gives -2.1 points with 90% CI $[-9.3, 5.0]$ over the full endpoint portfolio, and Kimi gives $+2.2$ points with CI $[-4.3, 8.7]$. These tranches are never pooled. A post-review exact-stratum audit shows why full-track equivalence should not be transported mechanically: A2 C3/C4 grounding each have G_0–N=+5 points with 90% intervals reaching +15, while the eight EIA held-out endpoints have -18.8 points with a wide interval. R14 therefore treats G_0–N as a surface-perturbation diagnostic. In the fresh A1 tranche, C3 and C4 grounding have G_0=N exactly and T exceeds both by +30 and +20

Question	Contrast	Status	Identified quantity
Repair original agent?	T–N	Executed	Incremental value of inserting the targeted procedure over the original agent.
Beat target-blind help?	T–G	Executed	Targeted-operation value relative to the frozen complexity-matched helper.
Does operation output matter on the same surface?	T–G_0	Executed	Same-surface operation-output contrast; G_0–N is reported separately as the placebo/surface perturbation.
Does integration surface matter here?	T–R_{surf}	Executed	Effect of placing the exact same precomputed operation output on the T surface versus ordinary context under the forced one-answer harness.

Table 4: Completed identification ladder. G_0 isolates operation output under the same helper surface without assuming stratum-level neutrality; R_{surf} isolates the integration surface while holding operation output exactly fixed.

points; EIA has T–N=+37.5 and T–G_0=+62.5, but G_0–N=-25, so the EIA net-repair claim is carried by T–N rather than a neutrality assumption.

R_{surf}: exact-output context-materialization control. R_{surf} runs the exact same targeted operation source as T on the same candidate evidence. Rather than exposing the output through `reusable_helper_output`, the harness adds the identical object as `retrieval_operation_output` inside the evidence package and exposes no callable helper. A zero-call parity audit verifies for every tested endpoint that (i) removing this one field recovers the original evidence package exactly, (ii) operation-output content hashes are identical between T and R_{surf}, and (iii) no candidate evidence is removed or reordered. On all 18 pre-frozen DeepSeek load-bearing endpoints, two fresh repeats per arm give T–$R_{\mathrm{surf}} = 0.0$ points with bootstrap 90% CI $[-5.6, 5.6]$ and post-hoc 95% sensitivity CI $[-8.3, 8.3]$. A paired-endpoint TOST at the prospectively frozen $\pm$10-point margin gives $p = 0.0121$ on each one-sided test; paired-t 95% CI is $[-8.5, 8.5]$. A deterministic parametric sensitivity calculation using the observed endpoint SD (17.15 points) gives about 53% probability of declaring equivalence when the true mean is exactly zero, so we do not characterize the design as high-powered. C3 grounding is T=R_{surf}=60% with five endpoint ties; C4 grounding is 70%/70% with one win, three ties, and one loss; EIA is ceiling at 100%/100%. The four strictly non-ceiling fresh endpoints have mean residual 0 but a wide 90% interval $[-25, 25]$. A secondary Kimi alignment check also ties on all runs. We therefore describe the observed DeepSeek portfolio as consistent with equivalence at the $\pm$10-point *portfolio-average* margin under this forced one-answer harness, with low sensitivity and a ceiling-heavy composition. Fine-grained non-ceiling effects, adaptive callable use, and independently designed temporal retrievers remain outside the claim.

Checkpointed execution and recovery. Every fresh G_0 and R model call is persisted before the next unit begins: an atomic raw JSON receipt, one fsynced CSV row, one fsynced JSONL row, and an atomic checkpoint summary keyed by endpoint$\times$repeat$\times$arm. Resume logic executes missing units only and never reruns a successfully checkpointed unit. Each experiment first runs a small runtime/protocol pilot whose promotion rule ignores scientific outcomes. This design was exercised in practice: transient control-channel failures occurred during the Kimi run while the remote process continued, and the already completed rows remained recoverable from CSV/raw checkpoints without replaying prior calls.

Every retained row binds endpoint, condition, repeat, skill hash, prompt hash, raw response hash, and provider response-ID hash. Kimi repeat 4 is excluded by hash provenance with no replacement. The exact TimeSage-EV replication remains separate from all source-native evidence.